

A Long Term Growth Perspective on Switzerland's Transition From The Golden Age To A Productivity Puzzle

Abstract

We examine Switzerland's economic growth using growth accounting and the Solow and Romer models. Decomposing GDP growth into capital, human-capital-adjusted labour, and TFP contributions across four sub-periods, we document a shift from TFP-led expansion during the post-war Golden Age to growth increasingly reliant on factor accumulation. The Solow model captures convergence toward a steady state, while the Romer framework better explains periods where technological progress drove TFP. The recent productivity puzzle (high R&D yet flat TFP) reflects the residual nature of TFP and the concentration of gains in a few multinational sectors, not a true deterioration in efficiency.

1 Growth Trends

1.1 Growth Trends and Growth Accounting

Trend lines fitted over Real GDP provide a good fit visually across different functional forms, particularly linear and quadratic forms. The trend line fits the linear form of Real GDP better than the log form and the same is true for the quadratic specification. Real GDP per capita has a flatter trend. As a robustness exercise, we also fitted trends in levels rather than logs (see appendix).

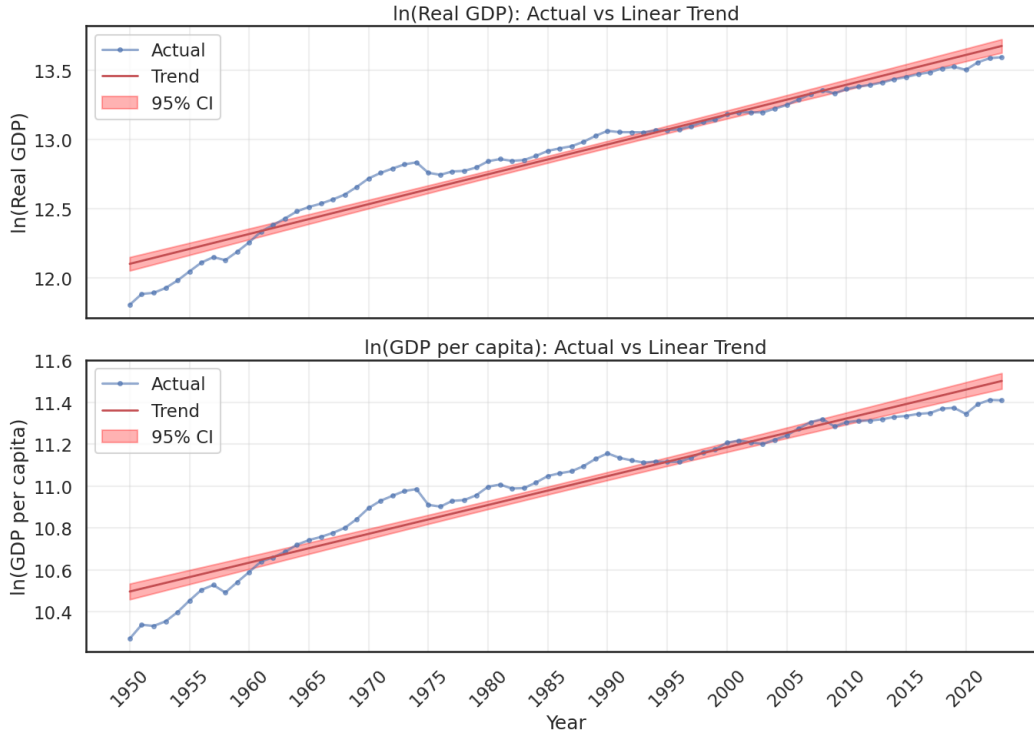


Figure 1: Linear trend time path for Log GDP and Log GDP per capita

Splitting the sample into sub-periods helps capture how growth has changed over time rather than relying on an overall average. Observing capital and labour trends, and looking for slope consistencies across variables, we arrive at four sub-periods: 1950–1974, 1975–1992, 1993–2006, and 2007–2023. Given the very limited fluctuations in the capital stock trend, we based this classification primarily on breaks in GDP and labour. These sub-period divisions coincide with significant transitions in Switzerland’s economy.

We decompose GDP growth using the human-capital adjusted production function:

$$Y = A \times K^{\alpha} \times (L \times HC)^{1-\alpha}$$

We use this specification because Human Capital data is available as an index (multiplying it with employment data provides a measure of effective labour). The three inputs are therefore TFP (A), capital stock (K), and effective labour ($L \times HC$)¹.

Table 1: Growth Decomposition

Period	Capital	Labour adjusted for Human Capital	TFP
1950–1974	29%	29%	42%
1975–1992	51%	69%	-20%
1993–2006	41%	29%	30%
2007–2023	38%	60%	1%

¹The benchmark capital share is $\alpha = 0.3$.

Several patterns stand out from the decomposition.

Period 1 (1950-1974):

TFP is the dominant contributor at 42%, with capital and effective labour contributing equally at 29% each. This is the only period where TFP leads growth.

Period 2 (1975-1992):

TFP drops sharply to -20% which is the lowest across all periods. Capital (51%) and effective labour (69%) together account for more than 100% of growth, meaning input accumulation had to more than compensate for the negative TFP just to deliver positive GDP growth.

Period 3 (1993-2006):

Capital leads at 41%, followed by a TFP rebound to 30%. Effective labour's contribution drops sharply from 69% to 29%. This is a more balanced growth pattern than in the previous period. We observe a significant recovery in productivity.

Period 4 (2007-2023):

TFP is at 1%. Growth is driven almost entirely by effective labour (60%) and capital (38%). This is the most input-dependent period alongside the second.

A few observations are worth noting

First, there is an inverse relationship between labour and TFP contributions across sub-periods. This partly suggests that periods of extensive labour growth tend to coincide with lower efficiency gains. Second, TFP is the most volatile component: it ranges from 42% to -20%, while capital remains in a narrower band (29%-51%). This volatility is partly because TFP, as a residual, absorbs any shock not captured by measured changes in capital and labour, a point we return to in Section 1.3.

1.2 Historical Context

The sub-periods identified in 1.1 align with historical events in Switzerland. We now connect the growth accounting patterns to these events.

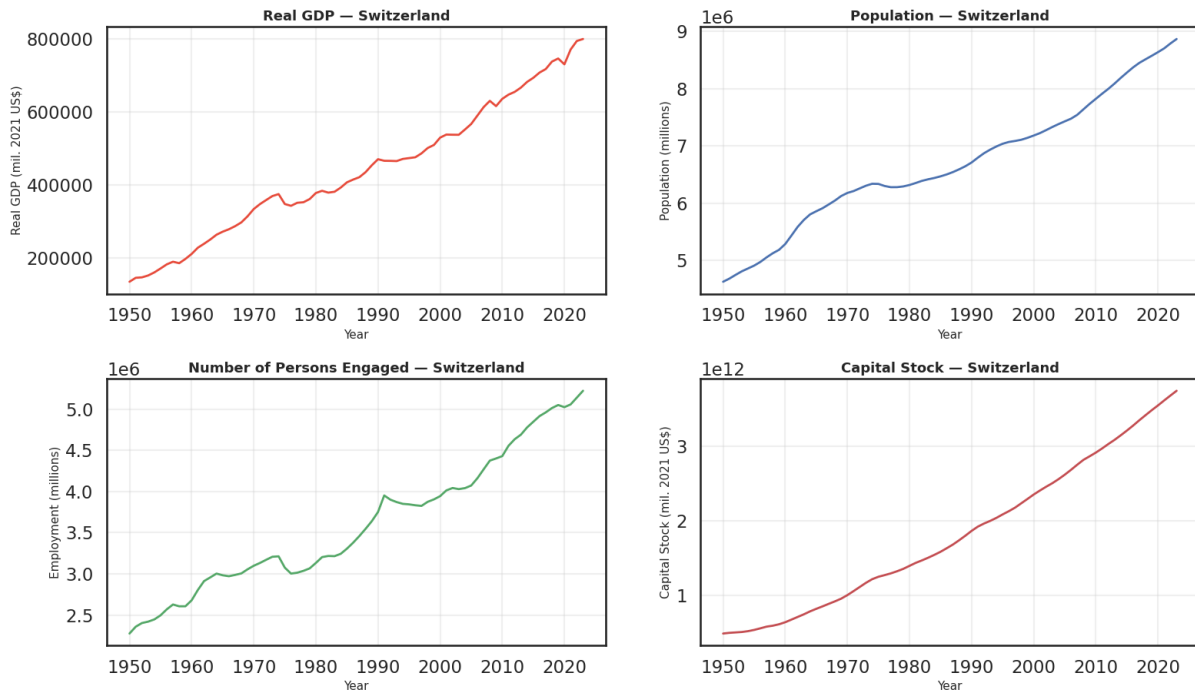


Figure 2: Time Series of Main variables

1950–1974: The Golden Age

This period, known as the 'Thirty Glorious Years', saw the highest average GDP growth in our sample (4.28%) Refer Table 2. Switzerland emerged from WWII with its institutions and physical capital intact. Following the Second World War, technological modernisation and structural transformation were key drivers of economic development (World Economic and Social Survey, 2017). Rising income is in line with the expansion of the middle class and consumption.

1975–1992: The Oil Shock Era

The post-war boom was followed by a massive storm. The 1973 OPEC oil embargo quadrupled oil prices, severely affecting energy-intensive industries. In 1979, the second major oil crisis took place following the Iranian revolution. These episodes had long-lasting effects on employment and income per capita, and help explain the -20% TFP contribution we observe. Growth in this period relied entirely on input accumulation. On the labour side, the high 69% contribution partly reflects the expansion of female workforce participation during these decades, which increased the number of employed people independent of the productivity slowdown.

1993–2006: Globalisation.

This period brings a breath of fresh air. The most notable driver was the globalisation boom (increasing trade openness, cross-border capital flows, and the birth of the internet). Switzerland, as a small open economy, benefited on two fronts: exporting high-value goods (pharmaceuticals, machinery, precision instruments) and adopting new technologies from abroad. This helps explain the TFP rebound to 30%. The combination of high capital and high TFP with low labour contribution tells us that growth in this period was explained by greater investment and improved efficiency, rather than adding workers. This was also the period that

Swiss watchmaking reinvented itself as a global luxury success (Donzé, 2014).

2007–2023: Shocks and Input-Dependent Growth.

This period experienced lower growth due to the 2008 financial crisis and the 2020 pandemic. Economic growth slowed significantly. Interestingly, the pandemic did not have as bad an effect as anticipated (World Economic Outlook, 2020), due to increased R&D investment which offset the effects of the pandemic. TFP is essentially flat at 1%, while labour contributes 60% (We will delve deeper into this seeming paradox later). Several factors help explain the high labour share. In 2008, Switzerland's retirement ages increased, expanding the workforce. Human capital also plays an important role: Switzerland continued investing in education throughout this period.

1.3 GDP and GDP per capita

GDP and GDP per capita both exhibit positive growth trends from 1950 to 2023. Switzerland is consistently one of the richest countries in the world by income per capita, and this status was maintained throughout the sample.

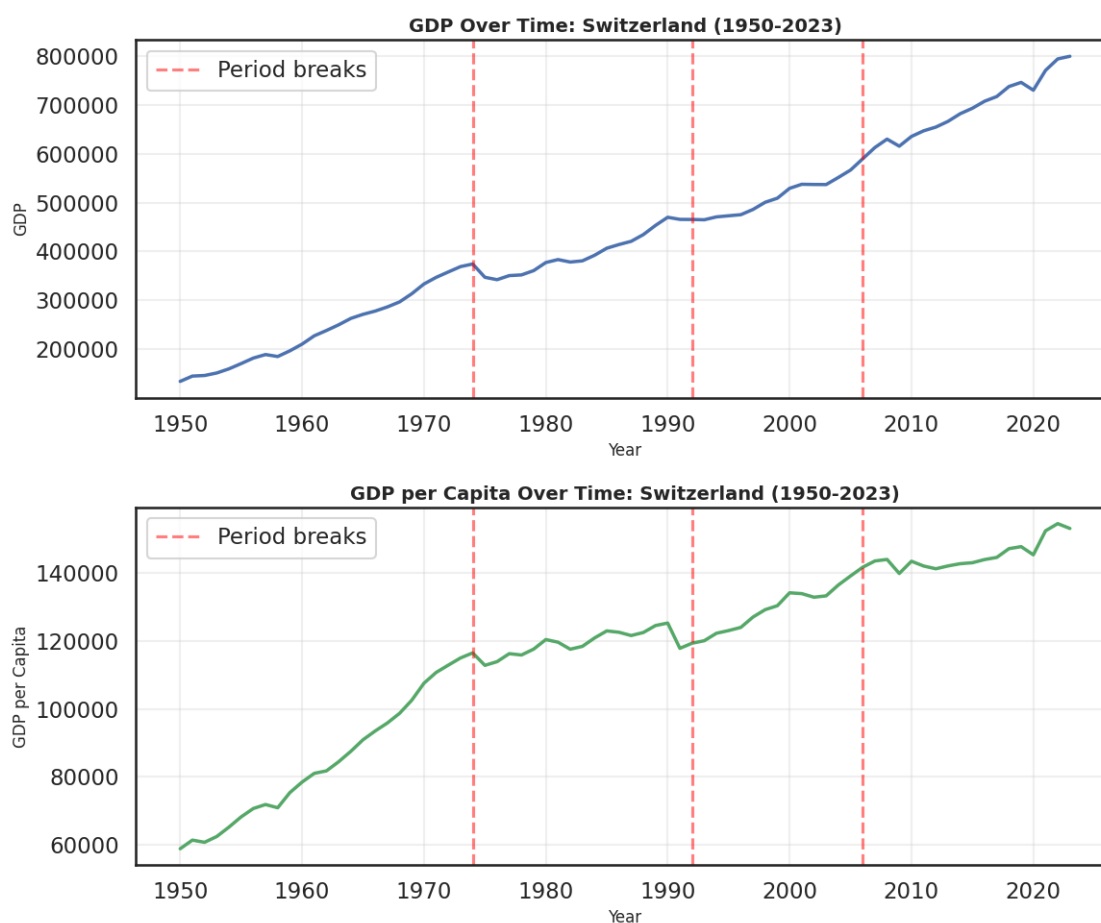


Figure 3: Time Series: GDP, period breaks

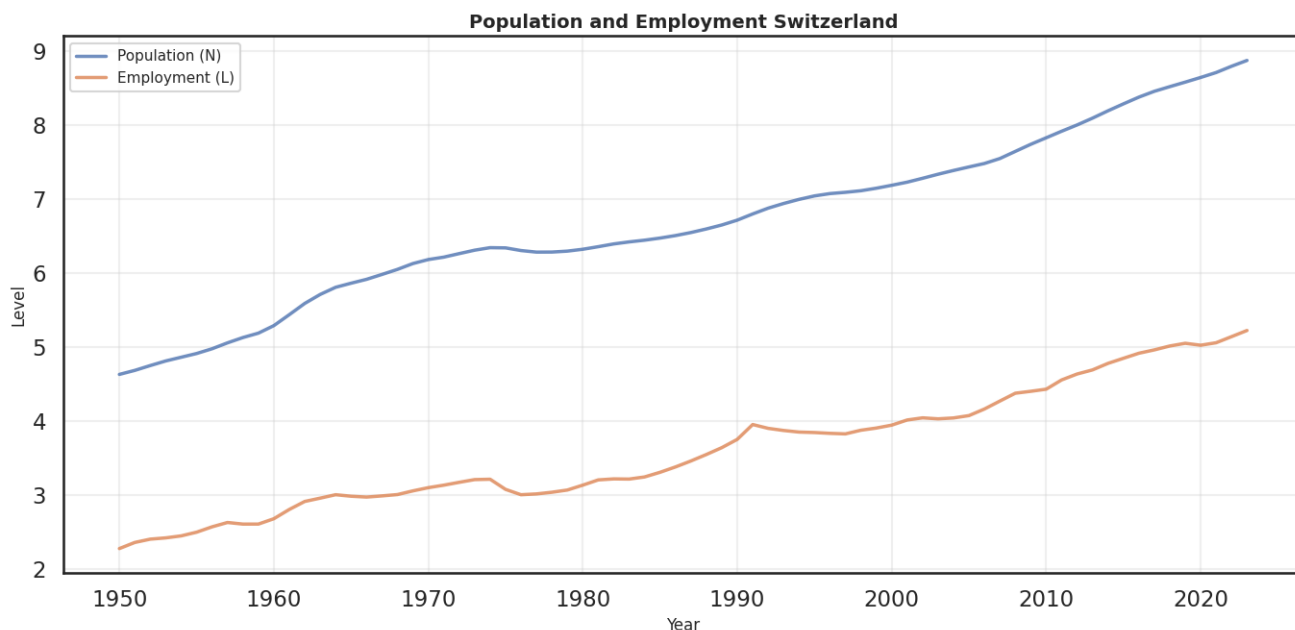


Figure 4: Time Series: Population (N) versus Employment (L)

Table 2: GDP Growth by Sub-Period

Sub-Period	GDP Growth (%)	GDP per Capita Growth (%)
1950–1974	4.28	2.85
1975–1992	1.73	0.33
1993–2006	1.83	1.27
2007–2023	1.66	0.40

GDP per capita growth consistently falls below overall GDP growth because population expansion reduces the gains measured on a per-person basis. To understand the role of population, we look at the trend lines of population and employment. The two are roughly parallel, which tells us that the employment rate has been roughly stable over time. This means that the labour contributions in our growth accounting largely reflect population expansion.

GDP per capita tracks TFP across sub-periods. In the first period, high GDP growth paired with high TFP (42%) delivered strong per capita gains. This means that productivity growth was being translated into improvements in living standards despite population expansion. In the second period, negative TFP pulled per capita growth down, even though input accumulation was positive. TFP rebounded in the third period, again contributing positively to GDP per capita growth. In the fourth period, with TFP essentially flat, per capita growth fell back to 0.40%.

This correlation between TFP and GDP per capita emphasises a key finding: sustained improvements in living standards seem to require productivity growth, not just more inputs. Input accumulation can keep GDP growing, but population growth dilutes those gains on a

per capita basis. Only when TFP is contributing does per capita income grow strongly.

2 The Solow Model

2.1 Solow and the Swiss Economy

We start with an intuitive assessment of whether the Solow model fits the Swiss economy, and then propose a more serious analysis comparing the growth explained by factor accumulation against actual growth.

Switzerland as a rich, mature economy

Switzerland has experienced positive long-run growth, but shifted from the early high-growth period to more modest rates recently. This pattern is consistent with the Solow model's prediction that an economy approaching its steady state should see declining growth rates. The marginal returns to capital accumulation diminish as the capital stock grows relative to effective labour. In that sense, Switzerland resembles an economy that has broadly reached the neighbourhood of its steady state.

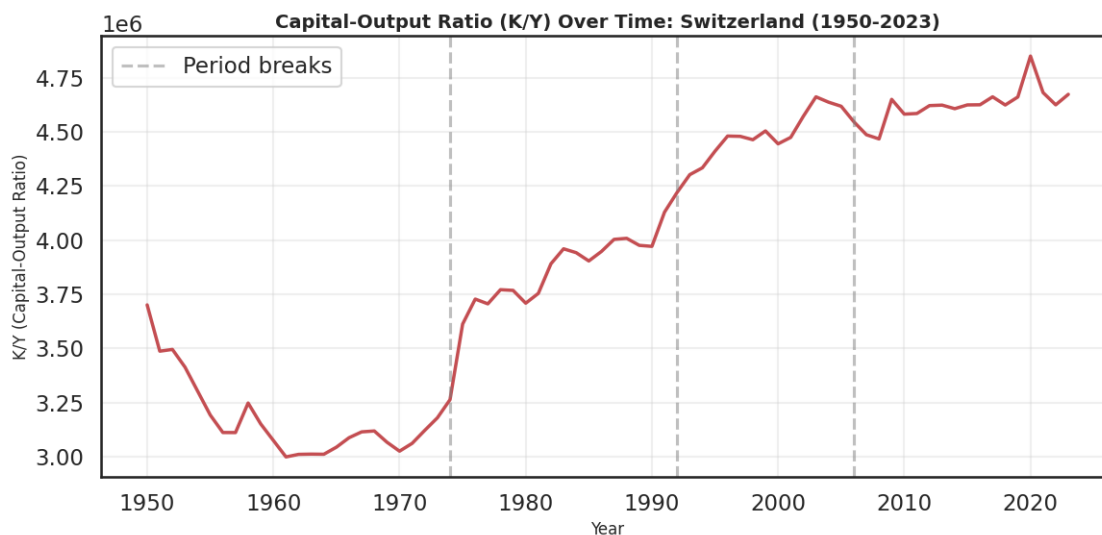


Figure 5: Time Series: Capital Output ratio, period breaks

The contribution of physical capital

The capital-output ratio (K/Y) has stabilised over the last two decades. In the Solow model, a constant K/Y is precisely the hallmark of an economy at its steady state since investment is enough to cover depreciation and keep up with effective labour growth, but not enough to push the capital intensity further.

The role of TFP from the Solow perspective

Conscious of our somewhat naïve initial interpretation, we now consider more seriously how our residual TFP measure informs the fit of the Solow model in both its versions: with and without technological progress.

2.2 Visual Analysis

The graphs below decompose the growth of output per effective worker into two components: the contribution from capital accumulation and the contribution from TFP. The red dashed line plots capital per effective worker (k). The red dashed line itself represents the capital per effective worker contribution (k^α), and the result is then indexed to a base year equal to 100². We then vary the starting year to better inform our analysis. This allows us to compare the theoretical growth driven by k against actual output growth.

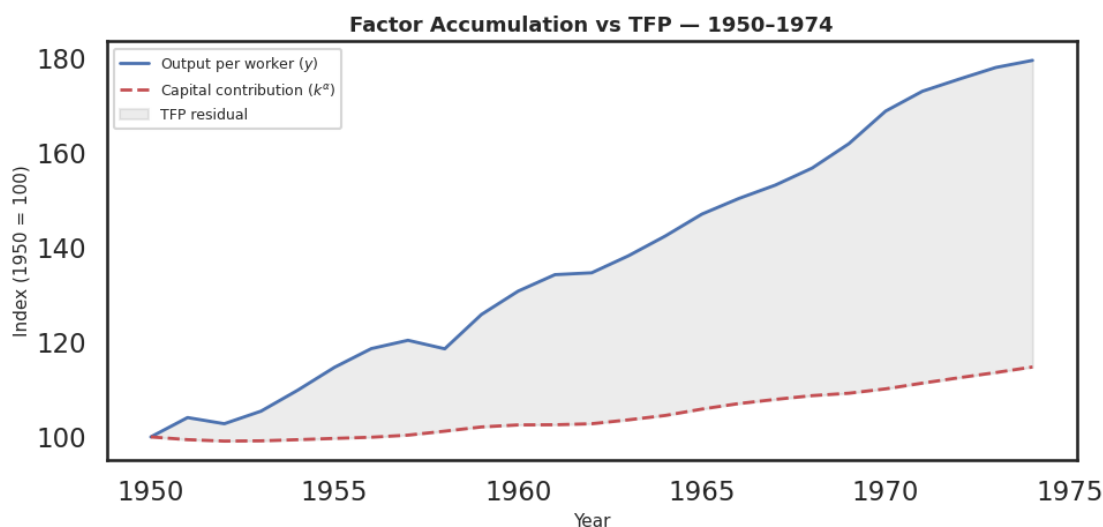


Figure 6: Factor accumulation versus TFP: 1950-74

First period (1950–1974)

Switzerland's growth during this period is highly consistent with a Solow model driven by technological progress. The solid blue line (actual output per effective worker) rises steeply and diverges significantly from the red dashed line (capital contribution). The widening grey area between the two indicates that output was growing much faster than what capital accumulation alone could explain. This large Solow residual is consistent with the "Golden Age" of European growth.

²More details about this calculation is available in the codes

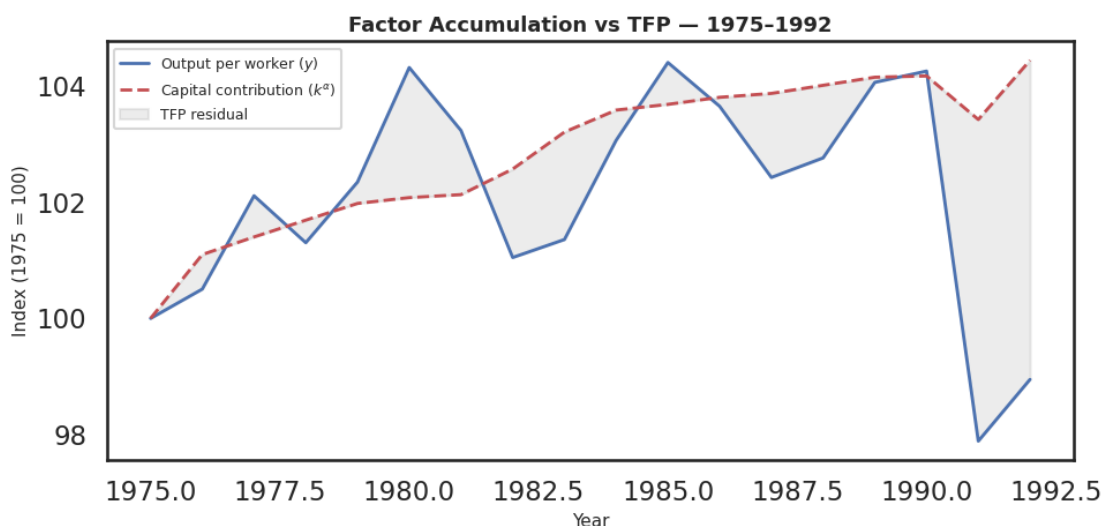


Figure 7: Factor accumulation versus TFP: 1975-92

Second period (1975–1992)

Output per effective worker fluctuates around the capital contribution line, with a contraction visible at the end. The economy was investing in capital per effective worker, but these investments did not yield the expected output growth in many years. This does not necessarily contradict the Solow model: it can accommodate TFP stagnation, since it treats technology as exogenous. However, the model is silent about why this happened.

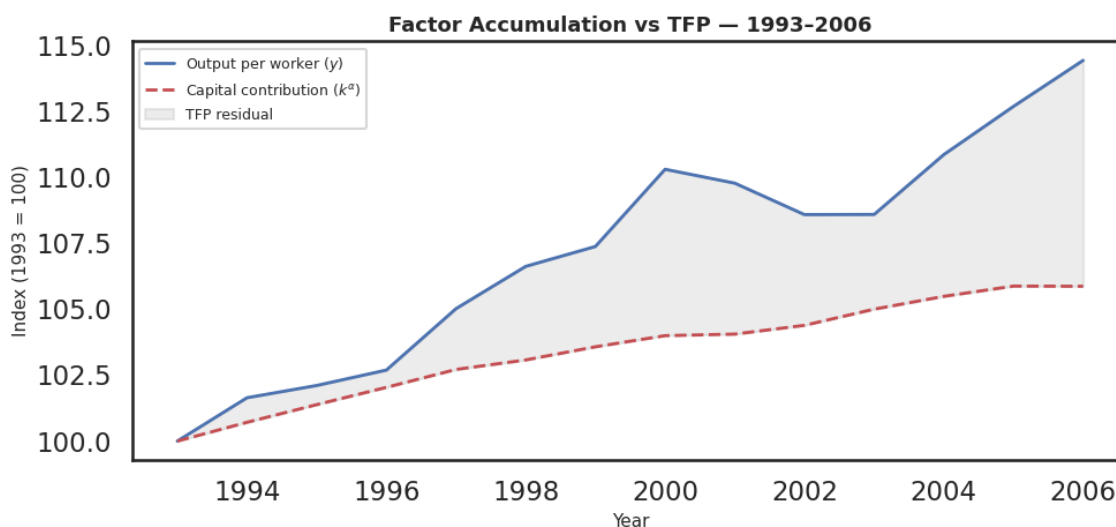


Figure 8: Factor accumulation versus TFP: 1993-2006

Third period (1993–2006)

Output per effective worker grows steadily above the capital contribution line, pointing to a positive and growing TFP residual. This is broadly in line with the increasing R&D and adoption of new technologies by firms during this period. We will revisit this topic in the context of the Romer model.

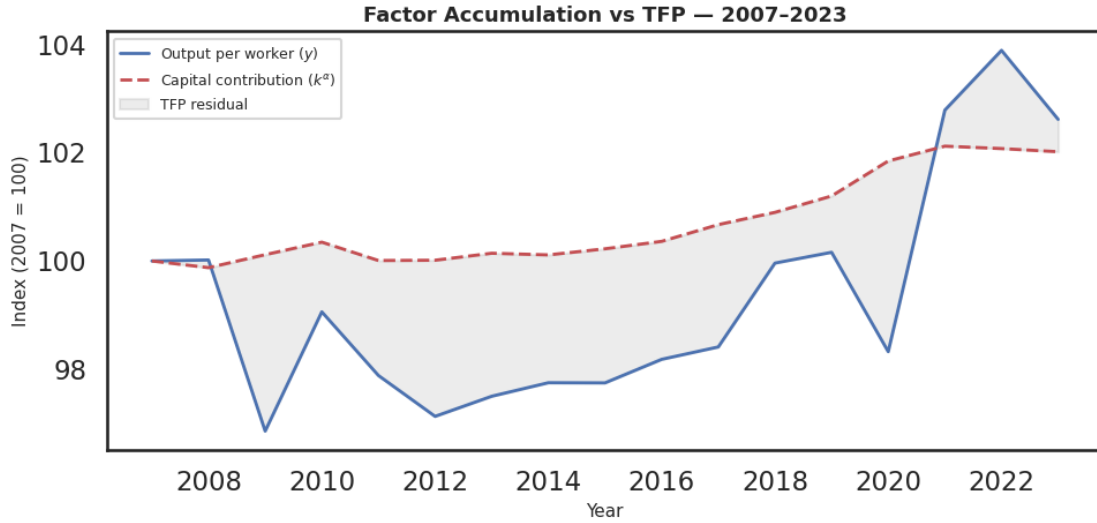


Figure 9: Factor accumulation versus TFP: 2007-23

Fourth period (2007–2023)

This period presents something of a puzzle. Our conclusions here are sensitive to the choice of base year: using 2007 shows a negative TFP contribution for much of the period, whereas using 2009 as the base would show a positive TFP contribution. This sensitivity is reasonable, since the period was hit by two major shocks (the Global Financial Crisis and the COVID-19 pandemic) that make the Solow residual noisier and harder to interpret. We argue that capital per effective worker appears to be the main driver of growth, which is a pattern consistent with an economy operating near its steady state in the Solow model without technological change. This idea will make more sense in the next section.

3 TFP and Labour Productivity

3.1 Theoretical Relationship

From the production function, labour productivity can be written as:

$$y = A \times k^{\alpha}$$

or

$$y = A \times k^{\alpha} \times HC^{1-\alpha}$$

Labour productivity depends on three components: TFP (A), capital per worker (K/L), and human capital (HC), each with its respective elasticity. The key implication is twofold. On one hand, an increase in TFP, holding other inputs constant, raises labour productivity. On the other hand, labour productivity can grow even if TFP is stagnant or declining, because capital deepening or human capital accumulation both raise output per worker independent of what happens to A .

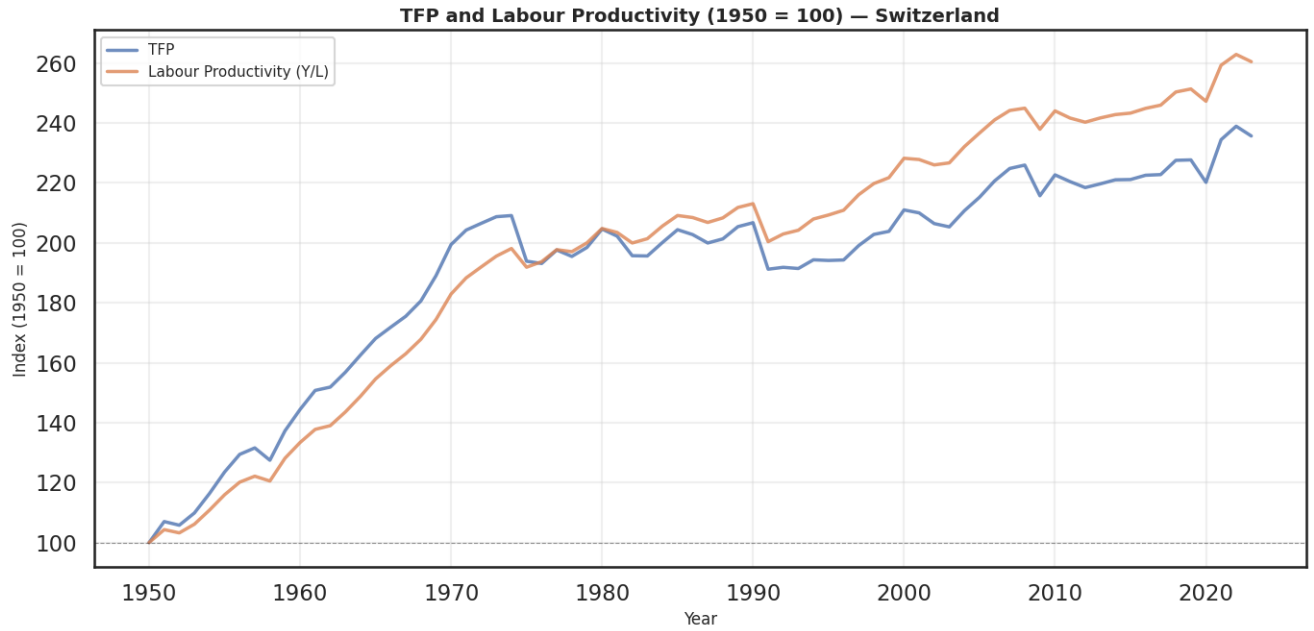


Figure 10: TFP and Labour Productivity

Neutral TFP - Solow Model

$$\begin{aligned}
 Y &= AK^\alpha L^{1-\alpha} \\
 \Rightarrow \frac{Y}{L} &= A \left(\frac{K}{L} \right)^\alpha \\
 \Rightarrow y &= Ak^\alpha \\
 \Rightarrow \frac{\partial y}{\partial A} &= k^\alpha > 0 \\
 \Rightarrow \frac{\partial^2 y}{\partial A^2} &= 0 \\
 \text{Elasticity: } \Rightarrow \frac{\partial y}{\partial A} \cdot \frac{A}{y} &= \frac{Ak^\alpha}{Ak^\alpha} = 1
 \end{aligned}$$

Neutral TFP, Human capital-augmenting Labour - Solow Model

$$\begin{aligned}
 Y &= AK^\alpha (HC \cdot L)^{1-\alpha} \\
 \Rightarrow \frac{Y}{L} &= A \left(\frac{K}{L} \right)^\alpha HC^{1-\alpha} \\
 \Rightarrow y &= Ak^\alpha \cdot HC^{1-\alpha} \\
 \Rightarrow \frac{\partial y}{\partial A} &= k^\alpha \cdot HC^{1-\alpha} > 0 \\
 \Rightarrow \frac{\partial^2 y}{\partial A^2} &= 0 \\
 \text{Elasticity: } \Rightarrow \frac{\partial y}{\partial A} \cdot \frac{A}{y} &= \frac{Ak^\alpha \cdot HC^{1-\alpha}}{Ak^\alpha \cdot HC^{1-\alpha}} = 1
 \end{aligned}$$

3.2 Relevance in Switzerland

Up to around 1970, both labour productivity and TFP grew broadly in tandem, suggesting that efficiency gains were the main driver of productivity growth during those decades. After 1970, however, the two diverged. Labour productivity (Y/L) rose consistently from 1950 to 2023. Growth is steeper in the early decades and flattens noticeably after the 1990s, but the overall trajectory remains positive throughout. TFP, by contrast, exhibits more volatility: strong growth in the 1950s & 1960s, a peak around the mid-1970s, followed by stagnation until 2000's.

The best way to communicate these patterns is to show both series on the same graph, indexed to a common base year (1950 = 100). This makes the divergence visible: the growing gap between the two lines represents the cumulative contribution of factor accumulation (capital deepening and human capital) to labour productivity beyond what TFP alone would predict. By plotting Y/L together with $Y/(HC \times L)$, we highlight how much of the observed productivity growth reflects improvements in human capital. For Switzerland, this comparison shows that education and training make a substantial contribution, particularly from the 1970s onwards when TFP growth slowed.

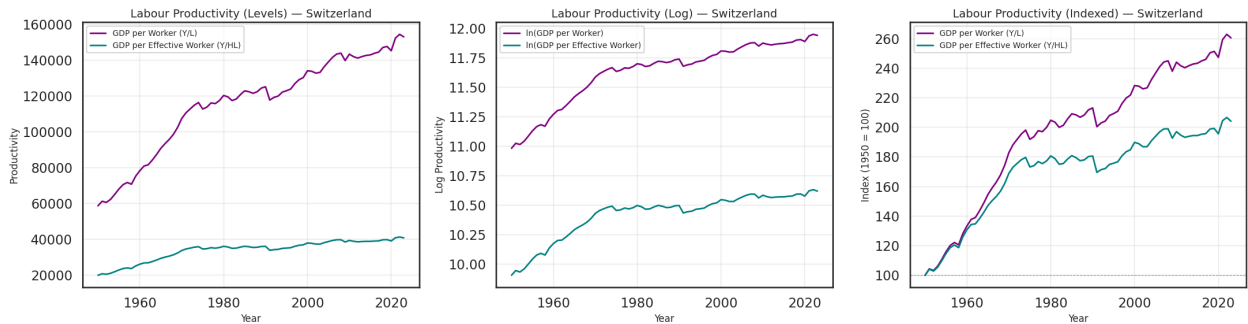


Figure 11: Time Series: Labour Productivity

3.3 Long-term divergence

The divergence between TFP and labour productivity in Switzerland reflects several reinforcing factors.

Factor accumulation as the dominant driver after 1970

Our growth accounting shows that capital and human-capital-adjusted labour became the main drivers of Swiss growth after 1970, while TFP contributed near-zero or even negatively in some sub-periods. Switzerland invested heavily in physical capital and education, which kept labour productivity rising even as TFP stagnated.

A dual economy

A recent IMF study (Zheng, 2025) documents that Switzerland's economy is essentially split into two. A few large multinational enterprises in pharmaceuticals, chemicals, and machinery drive most of the productivity gains, while the rest of the economy (particularly small

and medium enterprises and the services sector) has lagged significantly. Our aggregate TFP captures the average of these two realities,³ which helps explain why measured TFP growth appears modest despite Switzerland hosting some of the world’s most innovative firms.

4 The Romer Model

The Romer model endogenises technological progress through R&D investment by profit-seeking researchers in imperfect markets. Unlike the Solow model, in which technology is exogenously determined, long-run growth arises from allocating resources to idea production. Combined with an expanding researcher pool due to growing population, the non-rivalrous nature of ideas generates increasing returns to scale, resulting in sustained TFP growth, and in turn long-run economic growth in this model.

Switzerland, as an R&D intensive economy, is a natural candidate for examining the Romer model. As shown in Table 1, Switzerland exhibits Romer consistent growth with TFP as the dominant driver in the sub-period 1950-1974 (42%) and moderate in 1993-2006 (30%), while the sub-period 2007-2023 (1%) paints a complex picture. We examine each in turn.

4.1 Switzerland as an Innovation Economy

Switzerland, the fourth-highest investor in R&D worldwide, invests 3.4% of its GDP annually, surpassing the OECD average of 2.7%. With the private sector investing 2.3% of GDP, most of this funding went to R&D-intensive sectors⁴.

³As expected, our robustness analysis on α confirms that TFP pattern is not sensitive to the choice of capital share parameter, even though the levels differ. However, the dual Swiss economy implies that using a single $\alpha = 0.3$ is a strong assumption.

⁴The pharmaceutical and chemical industries (40%), metals and engineering industries (11%), research laboratories (13%), and new technologies (14%).

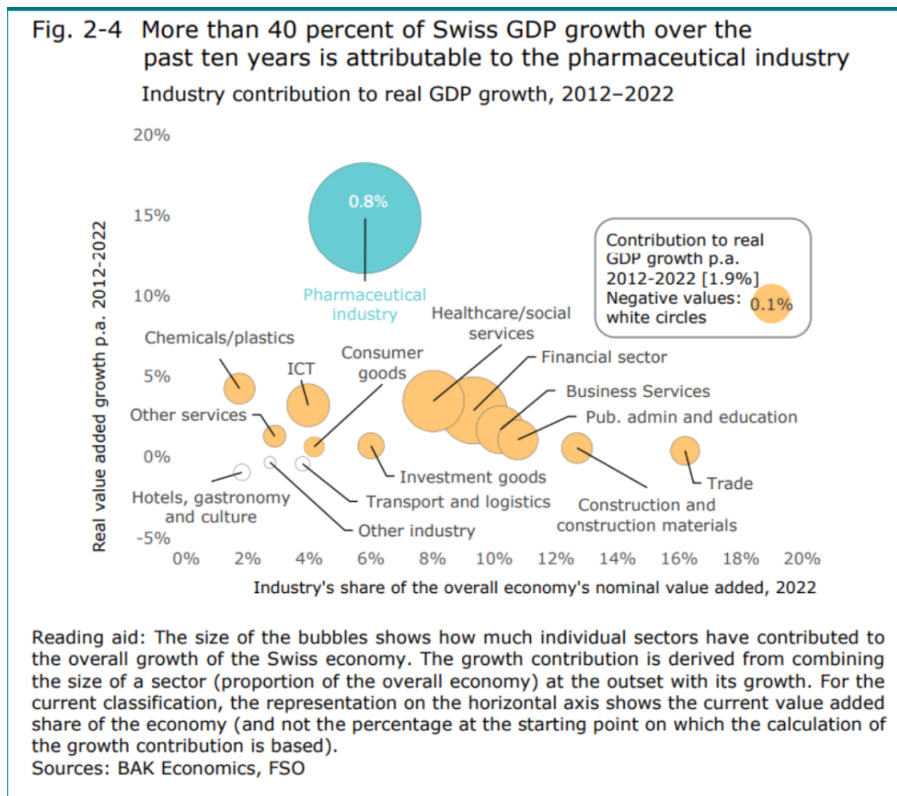


Figure 12: Industrial contributions to real GDP growth

Switzerland ranks first among 139 economies featured in the Global Innovation Index, which assesses their innovation capabilities. It leads internationally in patents, scientific publications and high-tech exports per capita, and demonstrates one of the highest innovation efficiency ratios globally.

4.2 Innovation-Driven Growth: 1950-1974

We observe the highest TFP contributions to GDP growth (42%) in this period, consistent with the Romer framework's prediction and argue that this TFP-driven growth reflects two mechanisms.

First, sustained growth in TFP during this period likely reflects a strong institutional framework established by 1950 which led to the post-war economic boom. Switzerland's institutional framework⁵ contributed to social infrastructure (Halls & Jones, 1999) that ensured the efficient allocation of resources and persistent productive activity. Crucially for the Romer mechanism, strong property rights and patent protection laws created excludability - key for incentivising R&D investment and production of ideas.

Second, Switzerland emerged from WWII with its institutions and physical capital intact, giving it a comparative advantage during the post-war reconstruction period resulting in the Golden age of growth. Growth accounting shows that TFP accounts for the majority of the

⁵rule of law, low corruption, strong property rights and patent protection, efficient markets

GDP growth while capital deepening plays a small role. Since labour productivity reflects both capital deepening and TFP, the strong alignment between Y/L and TFP suggests that efficiency gains were the main driver of productivity growth during this period (Figure 10).

The next sub-period 1975-1992, dominated by capital accumulation with negative TFP contributions, is better explained by the Solow framework as discussed above.

4.3 Globalization and Knowledge Spillovers: 1993-2006

Growth accounting for this period indicates that capital deepening accounts for the largest share in GDP growth (41%), while TFP makes a substantial contribution (30%). Positive population growth and high R&D expenditures suggests the existence of more researchers and ideas in the economy. However, as a small open economy, Switzerland's TFP growth in this period reflects not only domestic ideas production, but also global knowledge spillovers. The Romer model and its crucial assumption of a closed economy, can only partially explain this growth in TFP. Switzerland likely operates as both a frontier innovator in specific sectors (pharmaceuticals, watch-making) and an adapter of new technologies (IT) from other parts of the world.

4.4 The Innovation Paradox: 2007-2023

In contrast to the previous periods, the fourth subperiod provides a clear productivity puzzle - high R&D expenditures but low contributions of TFP. High R&D intensity paired with a high population suggests more researchers. At the same time, we also observe the fall of annual patent applications per million people in Switzerland. This divergence between rising research inputs and falling innovation outputs suggests congestion effects in idea production. Falling patent output, high growth rate of researchers and low TFP contribution can be explained through the 'stepping on toes' effect and a weakening 'giant shoulders' effect. Producing innovative ideas becomes harder when the existing stock of knowledge is large.

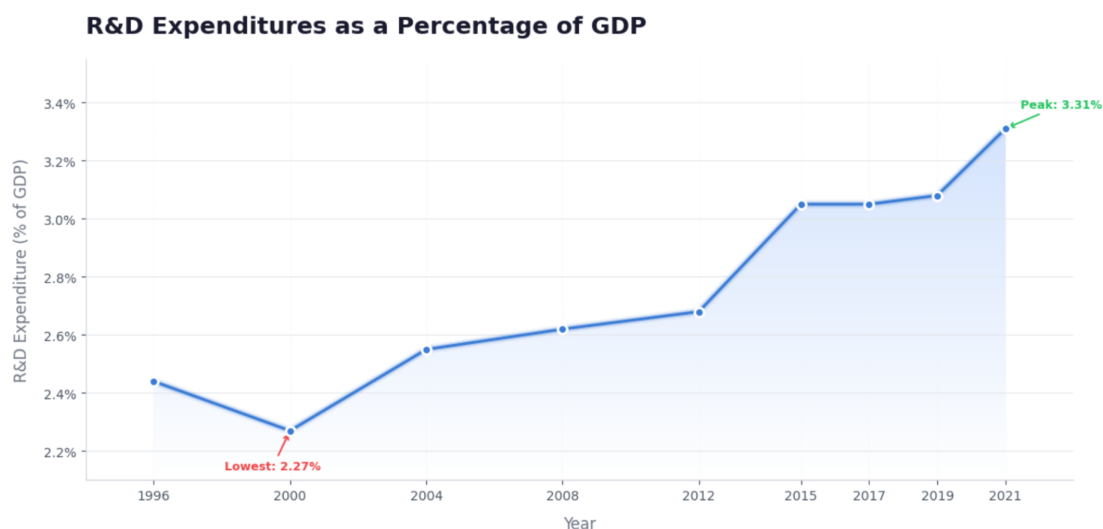


Figure 13: R&D Expenditures as % of GDP

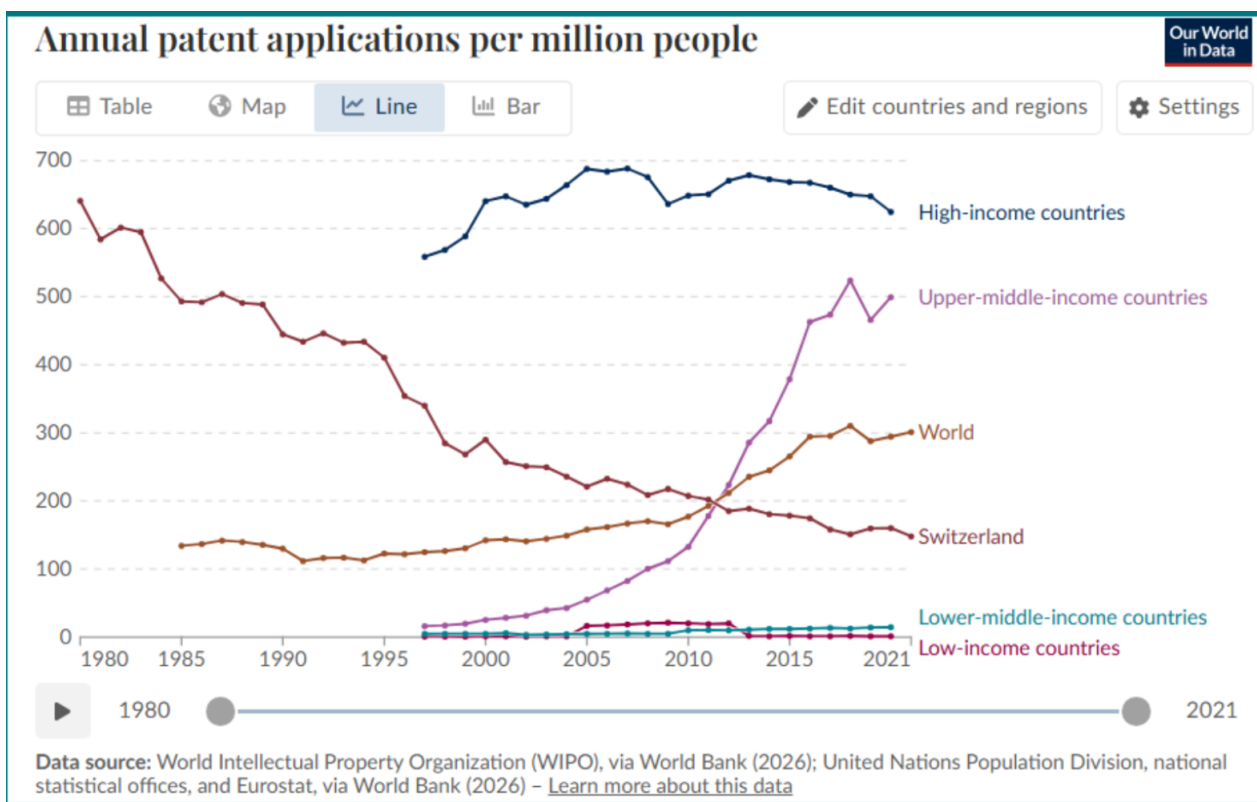


Figure 14: Annual patent applications: International comparison

This pattern seems to contradict the Romer model, which predicts that GDP growth should increase over time as the number of researchers and R&D investment rises. However, as we argued before, the low TFP contributions might be misleading. Since TFP is calculated as a residual, it is capturing the effects of two major negative macroeconomic shocks. Changing the starting year of the analysis would change the overall conclusion (see Section C).

Moreover, as already discussed, the most productive activities are becoming increasingly concentrated in just a few sectors (the “dual economy” issue in 3.3), which may also distort the aggregate picture.

5 Conclusion

Switzerland’s growth story is one of transition: from TFP-led expansion during the Golden Age to growth increasingly reliant on factor accumulation in recent decades. The Solow model captures the broad convergence toward a steady state, while the Romer framework better explains the innovation-driven periods. Neither model alone accounts for the full picture. The productivity puzzle of the most recent period (high R&D expenditure yet essentially flat TFP) reflects the residual nature of TFP, which absorbs the effects of major macroeconomic shocks, and the dual economy documented by the IMF, rather than a decline in efficiency gains itself. Sustained improvements in living standards ultimately require productivity growth.

References

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Appendix

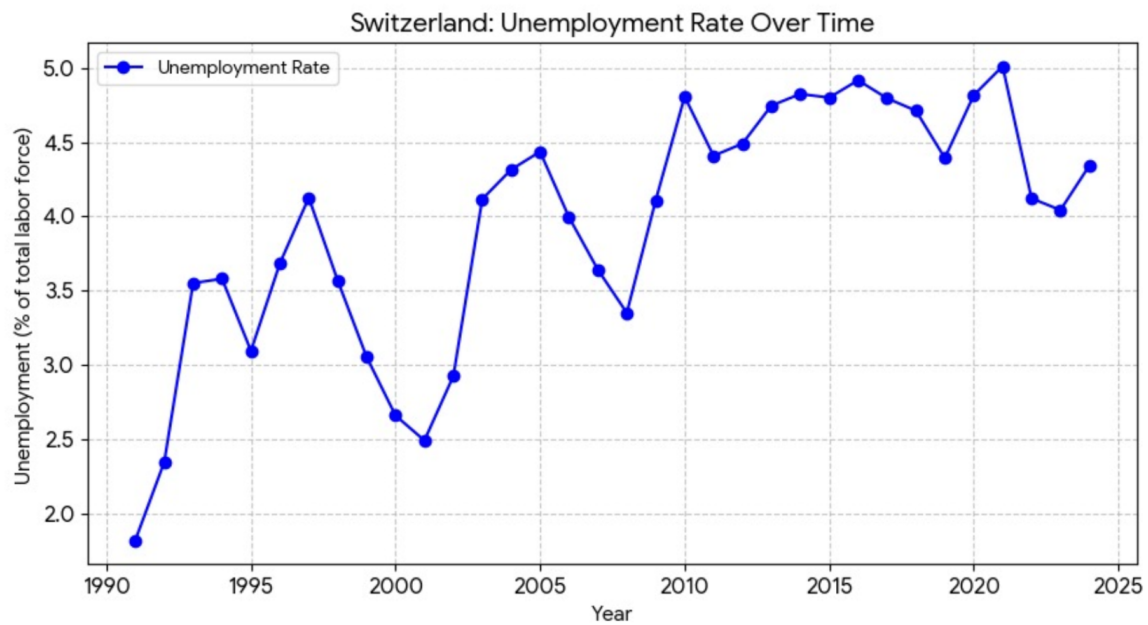


Figure 15: Unemployment rate over time

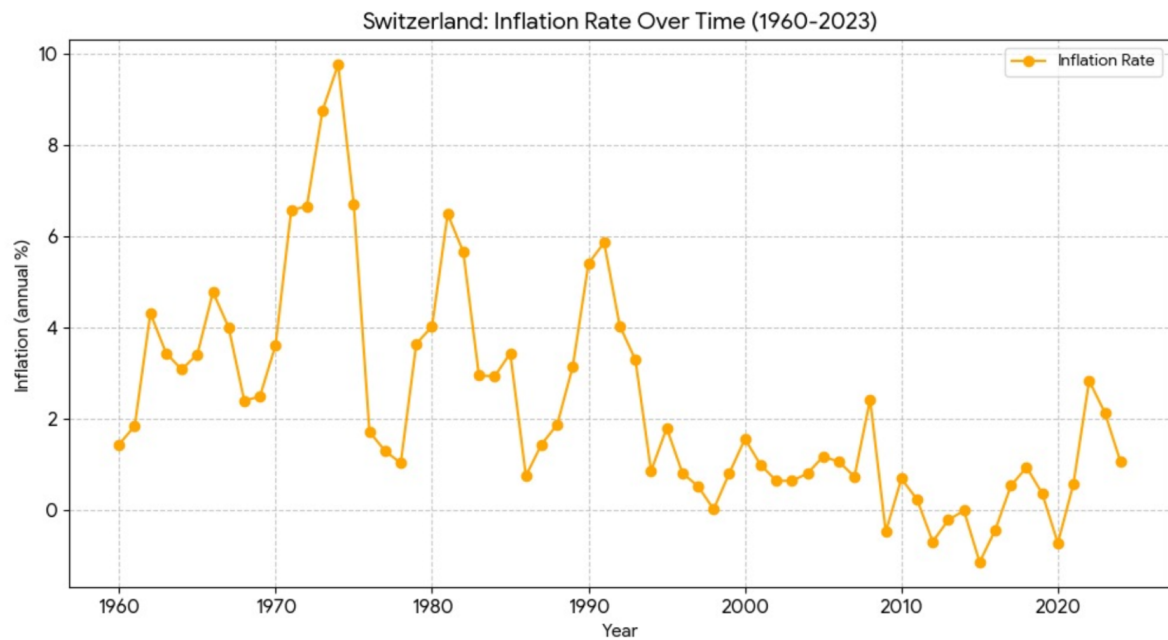


Figure 16: Inflation rate over time

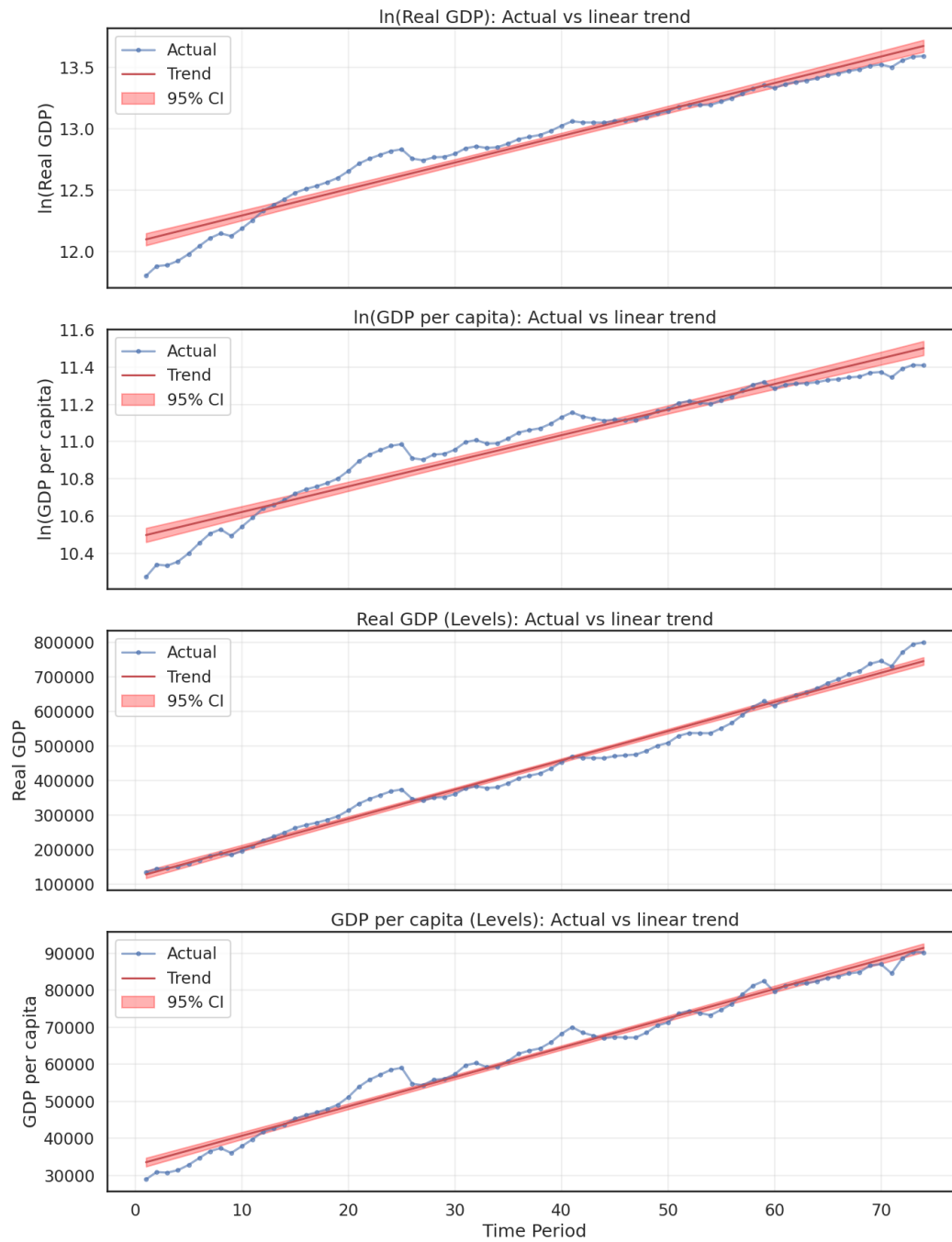


Figure 17: Fitting linear trends in GDP data

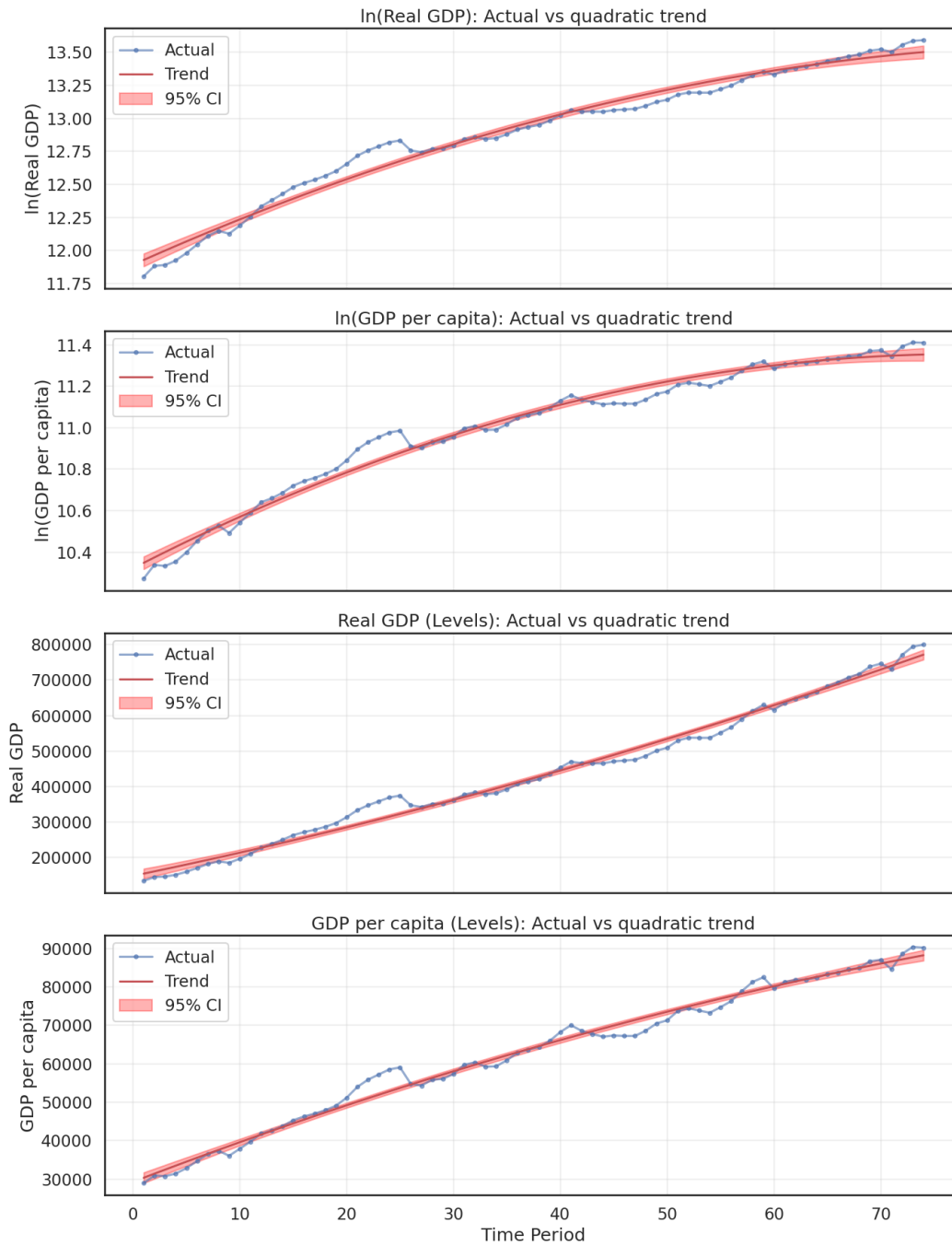


Figure 18: Fitting quadratic trends in GDP data

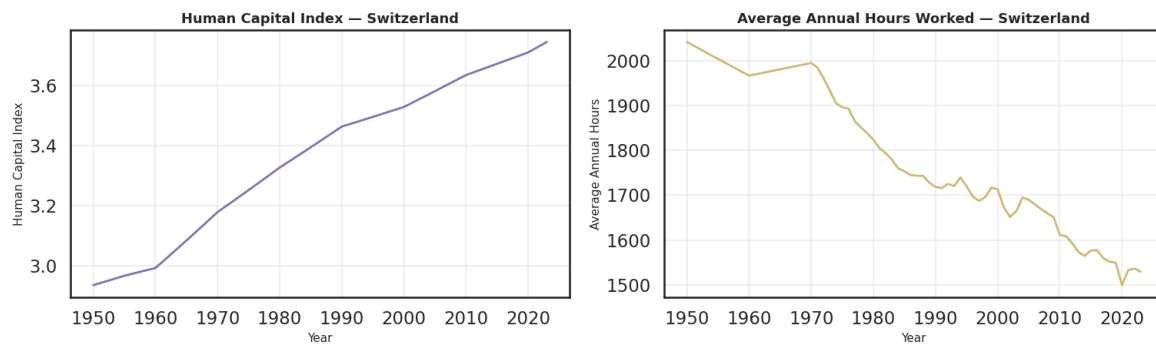


Figure 19: Time Series of main variables 1950-2023

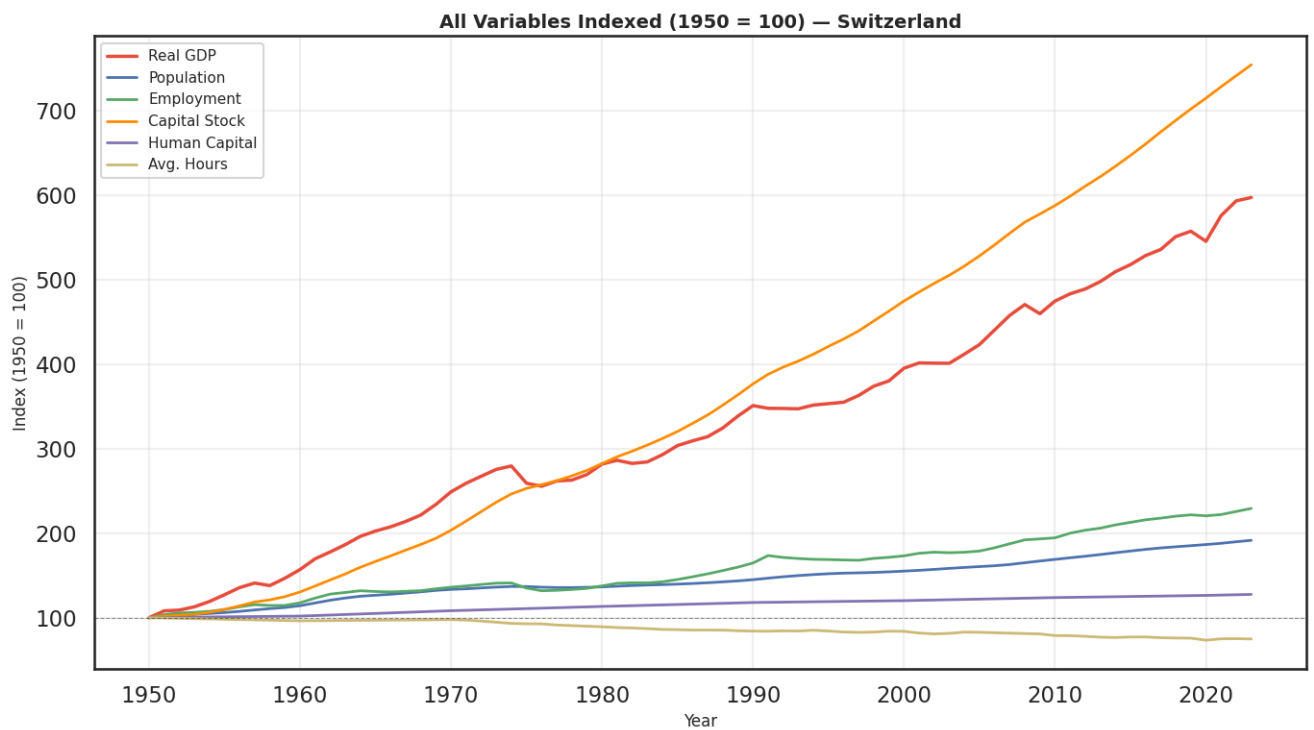


Figure 20: Indexed time series (main variables) 1950-2023

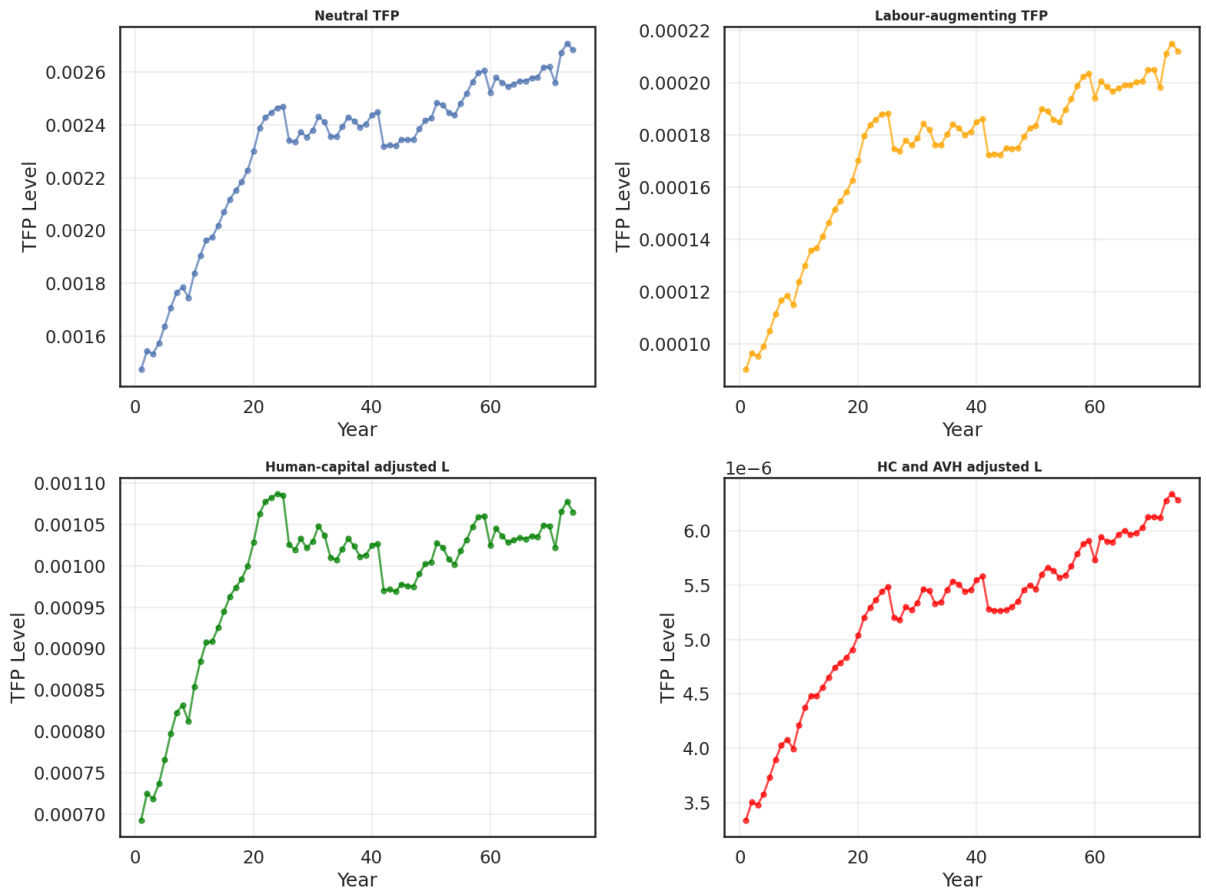


Figure 21: TFP calculated using different Solow Model specifications

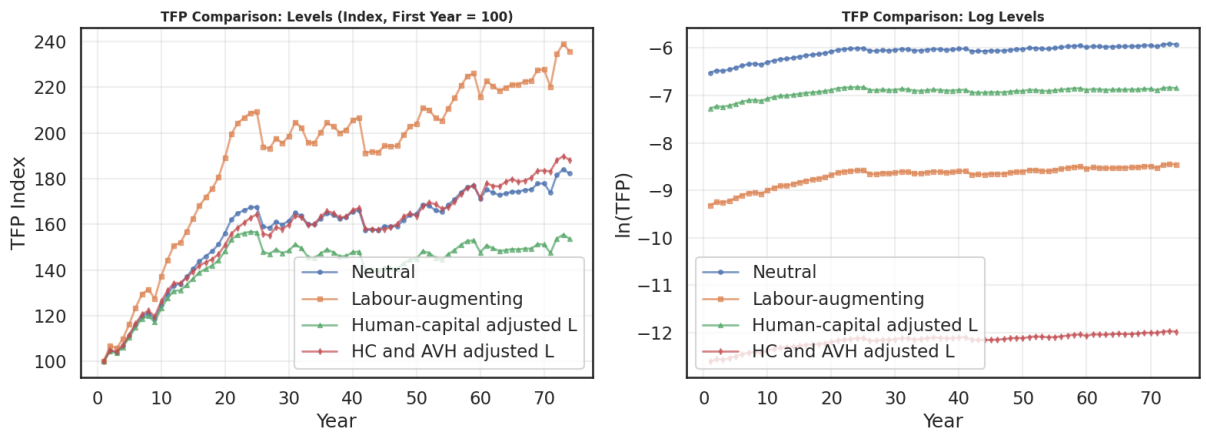


Figure 22: TFP calculated using different Solow Model specifications (comparison)

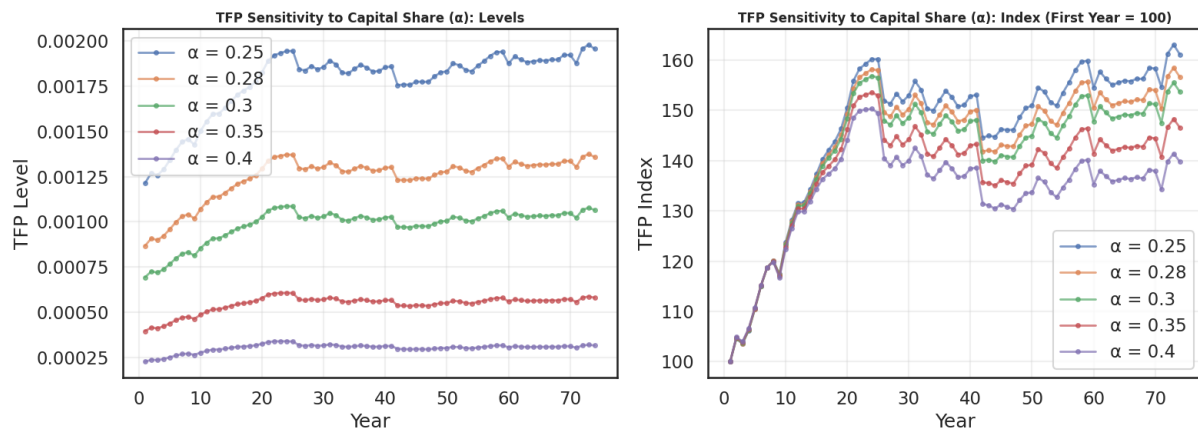


Figure 23: TFP robustness to alpha values

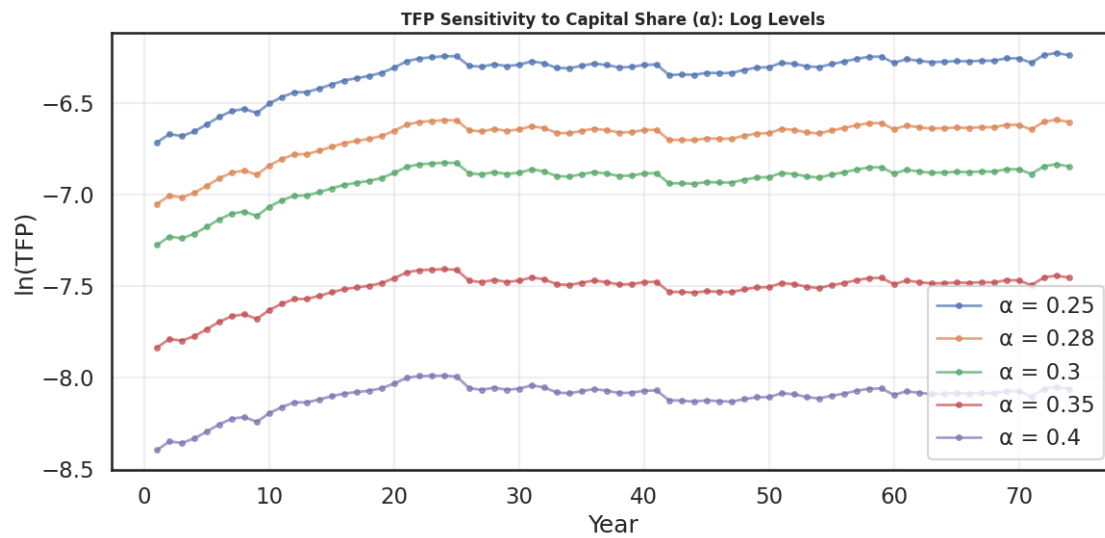


Figure 24: TFP robustness to alpha values (Log-levels)

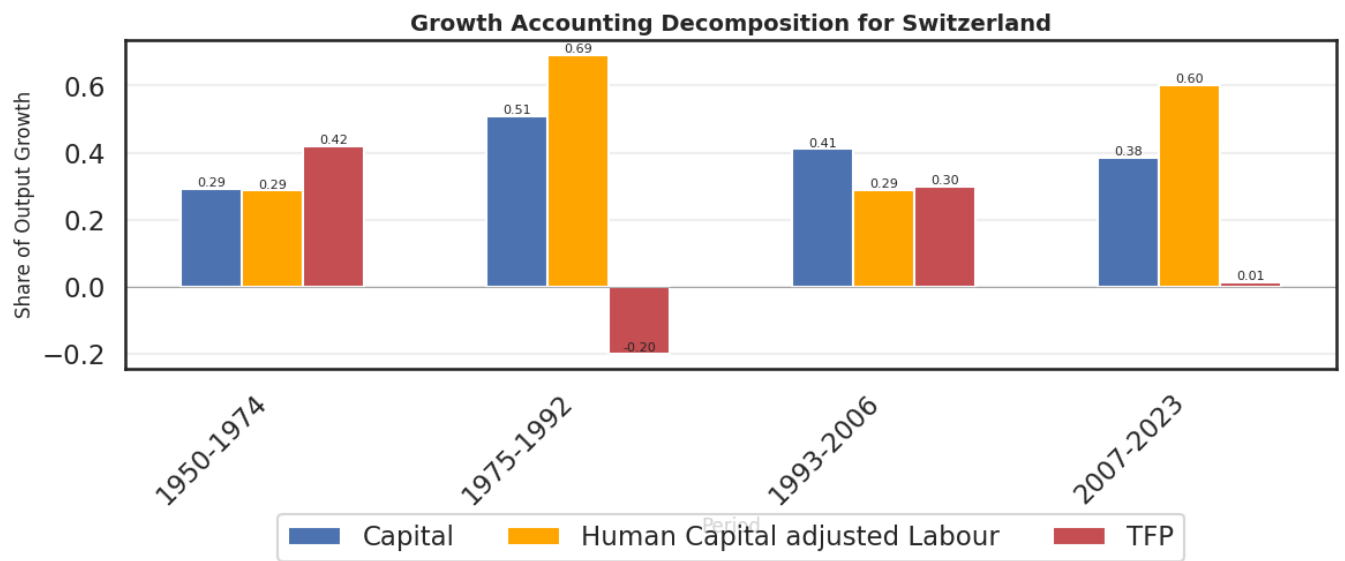


Figure 25: Growth accounting decomposition: $Y = A \times (K)^\alpha (HC \times L)^{(1-\alpha)}$

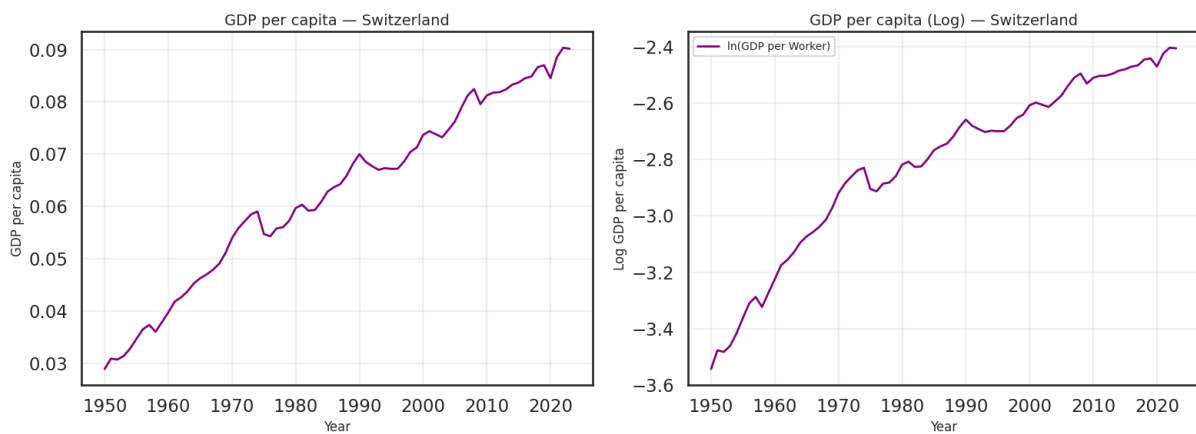


Figure 26: Time Series: GDP per capita

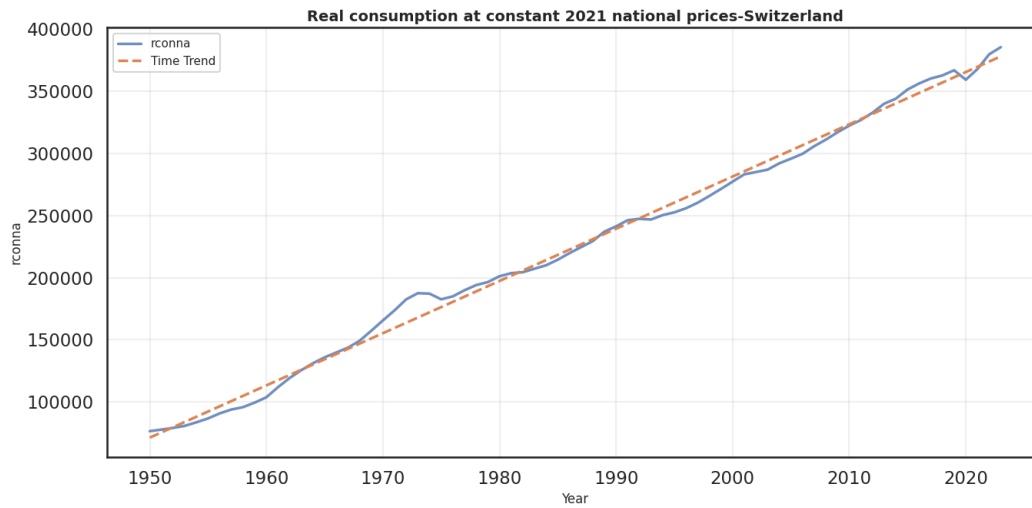


Figure 27: Time Series: Real Consumption

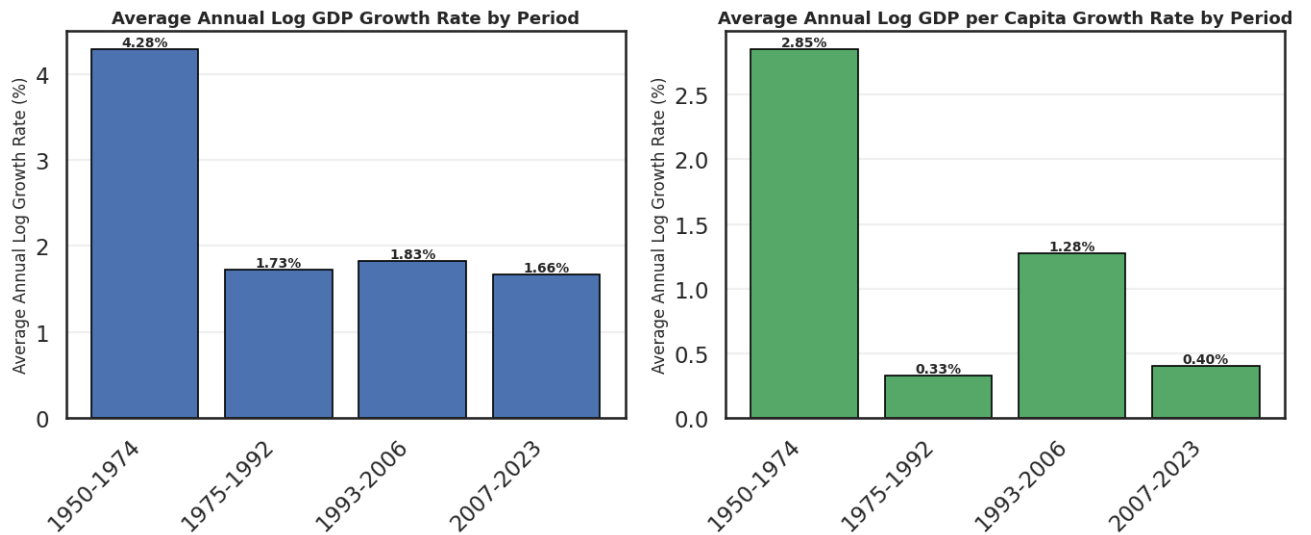


Figure 28: Average growth rates across sub-periods

Table 3: Growth Decomposition by Period

Period	Total Growth	K contrib	H contrib	A contrib	Sum
1950–1974	102.826497	0.292508	0.286827	0.420665	1.0
1975–1992	29.363015	0.510106	0.690688	-0.200794	1.0
1993–2006	23.793159	0.410802	0.288797	0.300401	1.0
2007–2023	26.618657	0.384383	0.602009	0.013608	1.0